

NetFRAME

Network Topology & Systems Reference

192.168.10.0/24 · km-cluster · 7 nodes · 3 GPUs · 4× RAIDZ2

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Document	NetFRAME Network Topology & Systems Reference
Version	1.0
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Classification	Public / Sanitized (service tags & serials omitted)
Method	Live discovery (nmap · Proxmox API · SSH facts · Headscale · ZFS) reconciled against the Home-Lab vault
Scope	Every host, VLAN, subnet, link, service, pool, and overlay path

Single-site production-grade home lab · Proxmox VE 9.2.3 · Juniper-cored VLAN fabric · ZFS + PBS · private GPU LLM platform

Contents

1	Executive Summary	2
2	Physical & Logical Topology	2
3	Layer 1 — Physical	3
3.1	Rack elevation (NetFRAME CS9000, 42U — rear panel removed for R730 depth) . . .	3
3.2	Key physical links	3
3.3	Power — split-bus one-line	3
4	Layer 2 — Switching & VLANs	4
4.1	Switch fabric	4
4.2	VLAN matrix	4
5	Layer 3 — Addressing, Routing, DNS	4
5.1	Static IP registry (VLAN 1)	4
5.2	Routing — per-node default gateway (live)	5
5.3	DNS & OOB / Servers VLANs	5
6	Overlay — Headscale Mesh VPN	5
7	Compute & Virtualization — km-cluster	5
7.1	Node hardware inventory	6
7.2	VM / LXC inventory	6
8	Storage Architecture	6
8.1	Randy ZFS datastore — 4× RAIDZ2 (36.7TB raw / ~23TB usable)	6
9	Services & Application Layer	7
9.1	Reverse proxy & published services (NPM, CT101)	7
10	AI / LLM Platform	7
11	Security Architecture	7
12	Data-Flow Walkthroughs	8
13	Observations, Drift & Recommendations	8
14	Failure-Domain / Blast-Radius	8
15	Appendix — Live Port / Service Scan	8

1 Executive Summary

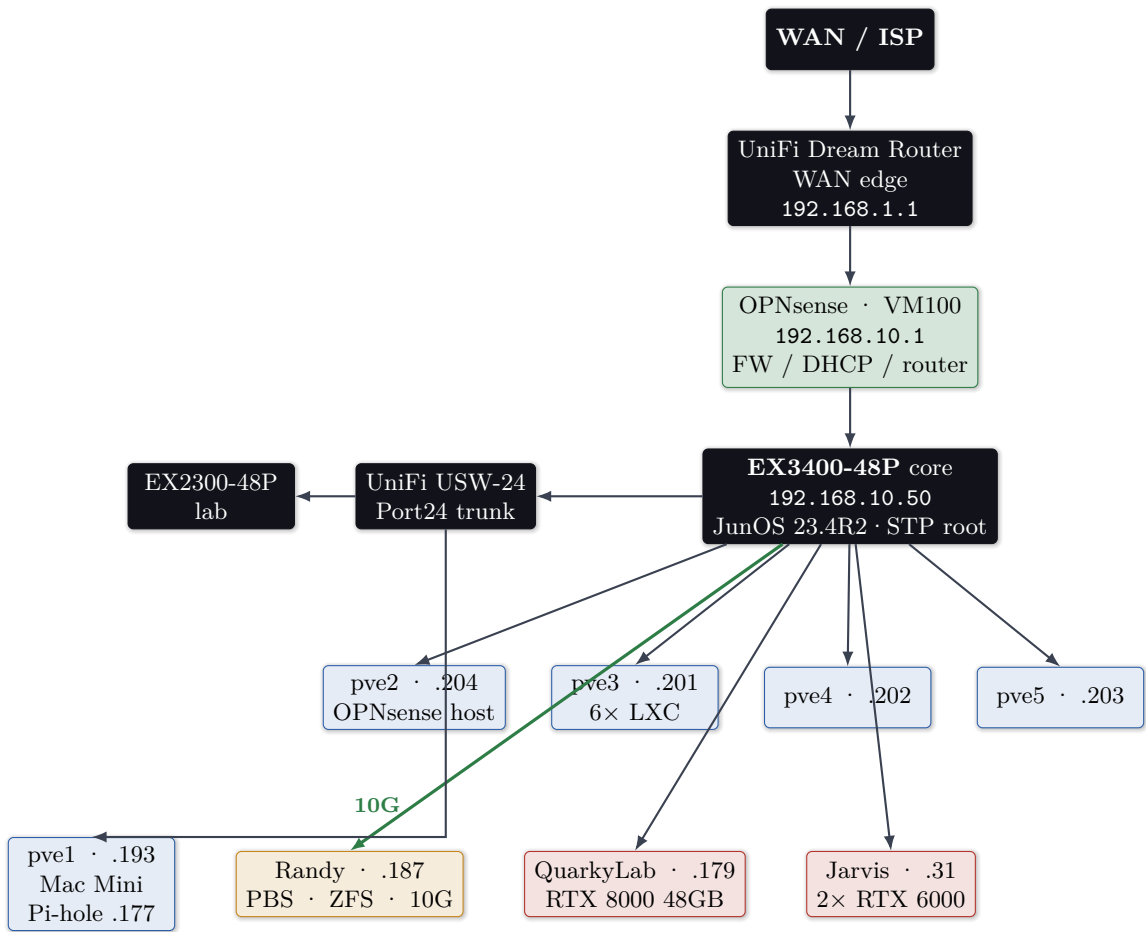
NetFRAME is a single-site, 42U home lab built around a **7-node Proxmox VE 9.2.3 cluster** (**km-cluster**), a **Juniper-cored, VLAN-segmented fabric**, and a **24-bay ZFS storage node** providing NFS and Proxmox Backup Server. It runs reverse-proxied web services with automated TLS, a Prometheus/Grafana/Loki observability stack, drive-health telemetry across ~50 disks, a Wazuh SIEM, an internal ACME PKI, a password vault, and a **GPU-accelerated private LLM platform** (RTX 8000 + dual RTX 6000, ~96 GB VRAM) with a custom OpenAI-compatible router that transparently escalates to the Claude API.

Proxmox nodes (quorate)	7/7	ZFS raw / usable	36.7 TB / ~23 TB
Standalone hypervisor	1 (pve1)	PBS datastore	24.6 TB
VMs / LXC	2 / 6	Monitored drives	~50
Managed switches	3	Overlay VPN nodes	9
VLANs defined	7	Aggregate RAM	~1.5 TB
Physical GPUs / VRAM	3 / ~96 GB	UPS (split-bus)	2 / ~2220 W

Edge model — deliberate double-NAT

A UniFi Dream Router is the WAN edge (192.168.1.0/24); **OPNsense** (VM 100 on pve2) is the LAN router / firewall / DHCP for 192.168.10.0/24 and the inter-VLAN gateway. The lab LAN therefore sits behind its own firewall boundary.

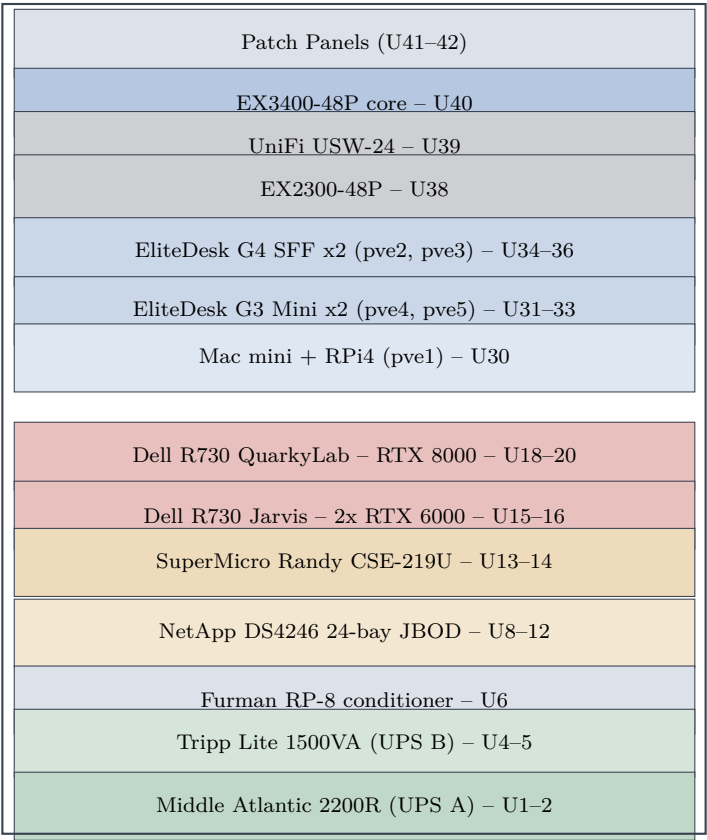
2 Physical & Logical Topology



Thick green = 10 GbE (Randy xe-0/2/0; Jarvis xe-0/2/2). All other links 1 GbE.

3 Layer 1 — Physical

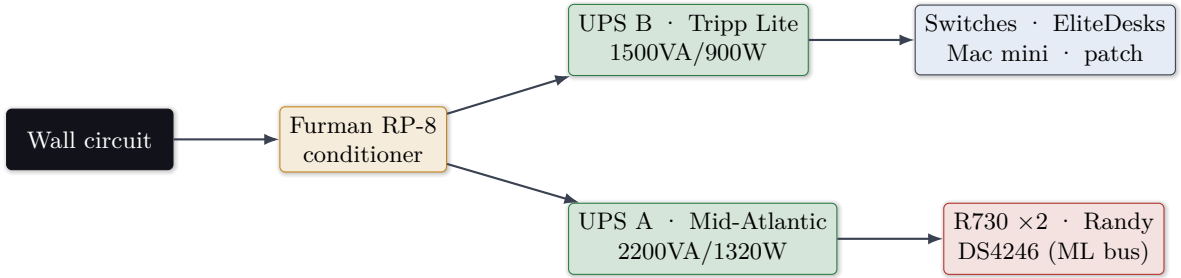
3.1 Rack elevation (NetFRAME CS9000, 42U — rear panel removed for R730 depth)



3.2 Key physical links

From	Port	To	Speed	Notes
EX3400	ge-0/0/46	UniFi USW-24 P24	1G	802.1Q trunk (native 1 + tagged 20/30/40/50/60/70)
EX3400	ge-0/0/45	EX2300	1G	Trunk, all VLANs
EX3400	xe-0/2/0	Randy nic3 (Mellanox)	10G	Storage / NFS data path
EX3400	xe-0/2/2	Jarvis enp132s0 (ConnectX)	10G	Access VLAN 30
EX3400	ge-0/0/24	QuarkyLab nic2	1G	Trunk native 1 + tagged 30
EX3400	ge-0/0/22	Jarvis nic2	1G	Native VLAN 1 (mgmt/corosync)
EX3400	ge-0/0/30·32·44	iDRAC/IPMI OOB	1G	Tagged VLAN 20 only
EX3400	xe-0/2/3	UniFi SFP2 DAC	—	DOWN — 10G/1G EEPROM mismatch

3.3 Power — split-bus one-line



Capacity watch

UPS A carries both GPU R730s. RTX 8000 alone can draw ~260 W under CUDA; with Jarvis’s 2× RTX 6000 online, full ML load approaches the 1320 W continuous rating. Both UPS units are exposed via **NUT 2.8.1** on pve3:3493 (UPS A over SNMP card 192.168.10.180, UPS B over USB).

4 Layer 2 — Switching & VLANs

4.1 Switch fabric

Device	Mgmt IP	OS	Role
EX3400-48P	192.168.10.50	JunOS 23.4R2-S7.4	Core — 48×1G PoE+, 4×10G SFP+, 2×40G QSFP+, dual PSU, STP root (prio 4096)
UniFi USW-24-250W	UniFi-managed	UniFi	Access / AP aggregation; Port 24 trunk
EX2300-48P	via trunk	JunOS	Secondary / lab isolation
UniFi Dream Router	192.168.1.1	UniFi	WAN edge (upstream of OPNsense)

4.2 VLAN matrix

VLAN 1	192.168.10.0/24	Management
VLAN 20	192.168.20.0/24	Trusted/OOB
VLAN 30	192.168.30.0/24	Servers
VLAN 40	192.168.40.0/24	IoT
VLAN 50	192.168.50.0/24	VoIP
VLAN 60	192.168.60.0/24	Guest
VLAN 70	192.168.70.0/24	Lab

Gateways are OPNsense SVIs (192.168.X.1); EX3400 carries the L3 mgmt SVI on irb.10.

JunOS ELS gotcha (live)

On this EX3400, `native-vlan-id` must be set at the **physical-interface** level, *not* under `unit 0 family ethernet-switching`. Misplacement previously caused a trunk outage. The tagged trunk to UniFi is `ge-0/0/46`.

5 Layer 3 — Addressing, Routing, DNS

5.1 Static IP registry (VLAN 1)

IP	Host	Role
.1	OPNsense	LAN gateway / FW / DHCP (VM100, pve2)
.2	UniFi USW-24	Access switch mgmt
.31	Jarvis	R730 LLM node
.50	EX3400	Core switch
.100/.199	Ares	Admin workstation (wired / WiFi)

IP	Host	Role
.148	Homepage	LXC 106
.177	Pi-hole	LXC 103 on pve1 (Mac Mini)
.179	QuarkyLab	R730 ML node
.180	UPS A SNMP	Mid-Atlantic OL2200R card
.181	Nginx Proxy Mgr	LXC 101
.182	Vaultwarden	LXC 102
.183	Grafana/Prom/Loki/Influx/Scrutiny	LXC 103
.184	Wazuh SIEM	VM 104 (QuarkyLab)
.185	Open WebUI	LXC 107
.186	Headscale	LXC 105
.187	Randy	PBS / ZFS / Jellyfin
.193	pve1	Mac Mini standalone (Pi-hole host)
.201–.204	pve3/pve4/pve5/pve2	Cluster nodes

5.2 Routing — per-node default gateway (live)

Node	Default GW	Iface	Note
pve2	192.168.10.1	vmbr1	correct (OPNsense)
pve3/4/5	192.168.1.1	vmbr0	onlink to UDR — off-subnet (finding F-1)
QuarkyLab	192.168.30.1	vmbr0.30	VLAN 30 egress
Jarvis	192.168.30.1	vmbr1	VLAN 30 egress (10G)
Randy	192.168.30.1	vmbr0.30	VLAN 30 egress

5.3 DNS & OOB / Servers VLANs

Pi-hole (192.168.10.177, FTL v6, LXC on pve1) is the LAN resolver + ad/tracker sink and serves *.netframe.local (e.g. llm./chat. → NPM .181). Public *.kylemason.org resolves via Cloudflare (grey-cloud) to NPM.

IP	Host	Note
192.168.20.20/21/22	QuarkyLab iDRAC / Jarvis iDRAC / Randy IPMI	OOB VLAN 20 (2026-07-03); credentials rotated
192.168.30.179/187/31	QuarkyLab / Randy / Jarvis	NFS / PBS / egress (dual-homed)

6 Overlay — Headscale Mesh VPN

Self-hosted Headscale v0.29.1 (LXC 105, 192.168.10.186:8080), tailnet 100.64.0.0/10.

Tailnet IP	Node	State
100.64.0.1 Ares (offline)	100.64.0.2 Randy	100.64.0.3 pve5
100.64.0.5 pve3	100.64.0.6 Jarvis	100.64.0.7 QuarkyLab
100.64.0.9 Fernanda-Mac	tag:ssh; all nodes online except laptops	100.64.0.8 pve1

7 Compute & Virtualization — km-cluster

7.1 Node hardware inventory

Node	Chassis	CPU	RAM	GPU	Kernel
pve2	EliteDesk 800 G4	i7-8700	31G	—	7.0.12-pve
pve3	EliteDesk 800 G4	i7-8700	46G	—	7.0.12-pve
pve4	EliteDesk 800 G3 Mini	i5-7500T	31G	—	7.0.12-pve
pve5	EliteDesk 800 G3 Mini	i5-7500T	31G	—	7.0.12-pve
QuarkyLab	Dell R730	2× E5-2699v4	503G	RTX 8000 48G	6.14.11-pve
Jarvis	Dell R730	2× E5-2687Wv4	377G	2× RTX 6000	6.14.11-pve
Randy	SuperMicro CSE-219U	2× E5-2690v4	125G	RX 580	7.0.12-pve
pve1	Mac Mini (2011)	—	—	—	standalone

Kernel pinning — do not upgrade

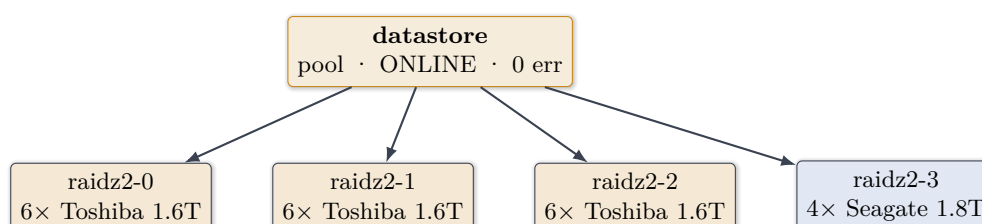
QuarkyLab and Jarvis are pinned to **6.14.11-9-pve** via `GRUB_DEFAULT`; 6.17+ breaks the NVIDIA 550.163.01 stack. Never run kernel upgrades or change GRUB default on either GPU node.

7.2 VM / LXC inventory

ID	Name	Host	Res	IP	Payload
VM100	OPNsense	pve2	4G	.1	LAN router/FW/DHCP v25.7
VM104	Wazuh	QuarkyLab	16G	.184	SIEM (no guest-agent — see §Security)
101	nginx-proxy	pve3	512M/8G	.181	Nginx Proxy Manager
102	vaultwarden	pve3	512M/10G	.182	Vaultwarden
103	grafana	pve3	2G/20G	.183	Grafana+Prom+Loki+Influx+Scrutiny
105	headscale	pve3	512M/4G	.186	Mesh VPN control
106	homepage	pve3	512M/8G	.148	Homepage + PeaNUT
107	openwebui	pve3	4G/16G	.185	Open WebUI

8 Storage Architecture

8.1 Randy ZFS datastore — 4× RAIDZ2 (36.7 TB raw / ~23 TB usable)



Boot: RAID-1 mirror (2× ST200FM0053 SAS) on **AVAGO 3108 MegaRAID**; data drives on a **separate LSI 9207-8e in IT mode**. PBS v4.2.2 (192.168.10.187:8007) exposes a 24.6 TB datastore.

Live finding — DS4246 dual-path SAS

The NetApp DS4246 shelf's 4 TB drives enumerate as **duplicate serial pairs** — dual-path SAS multipath through the shelf's two I/O modules presenting each disk on two paths. Configure **multipath/dm** before pooling. (This also flags drift: docs list 2 TB drives; live shelf is 4 TB-class, buildout in progress.)

PBS storage pin (F-4, resolved)

The VLAN-30 migration silently reprinted PBS storage to .30.187 and stalled VLAN-1 node backups from 2026-07-02; fixed 2026-07-06 back to 192.168.10.187. **Keep PBS storage on .10.187.**

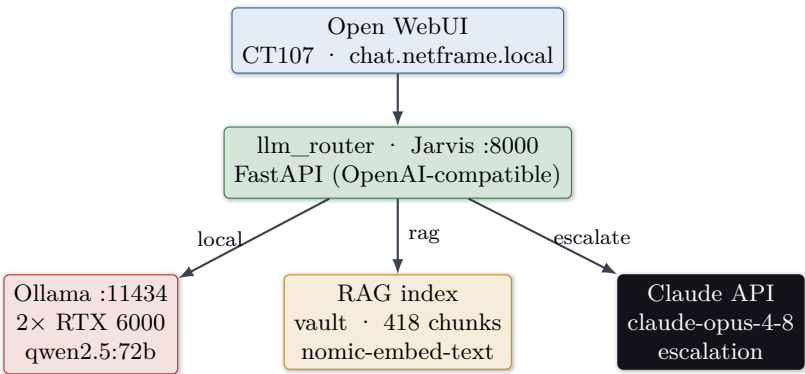
9 Services & Application Layer

9.1 Reverse proxy & published services (NPM, CT101)

Hostname	Backend	TLS	Auth
vault.kylemason.org	.182	CF DNS-01	Vaultwarden
grafana.kylemason.org	.183:3000	CF DNS-01	Grafana
homepage.kylemason.org	.148:3000	CF DNS-01	Basic (kyle)
wazuh.kylemason.org	.184:443	CF DNS-01	Wazuh
chat.netframe.local	Open WebUI .185	internal	admin
llm.netframe.local	Jarvis .31:8000	HTTP	Pi-hole-scoped

Observability: Prometheus (.183:9090, localhost-only, 8 exporters), Grafana v13, Loki, InfluxDB, Scrutiny (~50 drives, collectors on Randy/QuarkyLab/Jarvis), NUT (pve3:3493), CrowdSec IPS, step-ca PKI (pve2:443).

10 AI / LLM Platform



On-prem, cited answers over your own notes

The "rag" model grounds answers on the Home-Lab Obsidian vault (418 chunks, cosine index) with [source] citations — served entirely on GPU on Jarvis, with a one-flag escalation to Claude Opus for hard queries. QuarkyLab’s RTX 8000 48 GB drives a parallel DUNE research RAG agent (Fernanda).

11 Security Architecture

- **OOB isolation (Phase 1):** all three BMCs moved off flat VLAN 1 onto tagged VLAN 20; default credentials rotated to Vaultwarden; reachable only from Ares **enp0s31f6.20**.
- **Servers VLAN 30:** GPU/storage nodes dual-homed (NFS/PBS/egress on 30; corosync/mgmt on 1).
- **Ingress chokepoint:** all web traffic via NPM; admin planes localhost- or Ares-bound (pentest F-03/F-05, verified).
- **Defense in depth:** Wazuh SIEM + CrowdSec IPS + step-ca PKI + Vaultwarden secrets.

Operational — Wazuh guest agent (F-5)

VM 104 lacks `qemu-guest-agent`; a QuarkyLab host reboot hard-stops it and `wazuh-indexer` returns unhealthy. Recovery: `qm stop 104 && qm start 104`, wait ~4 min (healthy = dashboard 302→/login).

12 Data-Flow Walkthroughs

A · Grounded chat: Browser → Pi-hole resolves `chat.netframe.local` → NPM `.181` → Open WebUI `.185` → `llm_router` Jarvis `.31:8000 (rag)` → embed + cosine search vault → Ollama `qwen2.5:72b` tensor-split → cited answer.

B · Nightly backup: `pvescheduler` → `vzdump` (02:00 CT / 03:00 VM) → PBS `192.168.10.187:8007 (VLAN 1)` → prune 7d+4w.

C · Storage: GPU nodes mount Randy NFS over **VLAN 30 10G** (`.30.187, xe-0/2/0`); `corosync` stays on VLAN 1.

D · BMC access: Ares tags VLAN 20 (`enp0s31f6.20`) → EX3400 `ge-0/0/30/32/44` → iDRAC `192.168.20.20-22`.

13 Observations, Drift & Recommendations

#	Sev	Finding	Recommendation
F-1	Med	Inconsistent default GWs: <code>pve3/4/5</code> via <code>192.168.1.1</code> (off-subnet, bypasses OPNsense)	Repoint to <code>192.168.10.1</code> or document intent
F-2	Low	Ares wired mgmt leg link-down; VLAN 20 reroutes via WiFi	Restore wired leg; verify <code>ip route get</code>
F-3	Med	DS4246 drift + dual-path duplicates	Configure multipath before pooling; update docs
F-4	✓	PBS storage was repointed to <code>.30.187</code> (broke back-ups)	Fixed; keep on <code>.10.187</code> ; monitor backup age
F-5	Low	Wazuh VM lacks guest-agent	Install <code>qemu-guest-agent</code> ; cold start
F-6	Low	<code>xe-0/2/3</code> DAC down (EEPROM mismatch)	Replace SFP or decommission
F-7	Info	VLANs 40/50/60/70 lightly populated	Populate or prune
F-8	Low	Amazon Echo (Alexa) on flat mgmt VLAN 1 (<code>.112</code>); <code>:1080</code> SOCKS5 advertises no-auth but does not relay (protocol-verified) — benign firmware artifact	Move to IoT VLAN 40 (egress-only); scheduled to move soon

14 Failure-Domain / Blast-Radius

Component	If it fails	Mitigation
<code>pve2</code> (OPNsense)	No LAN routing/DHCP/inter-VLAN — whole LAN	onboot; serial-console runbook
<code>pve3</code> (6 LXC's)	NPM/Vault/Grafana/Homepage/Headscale/OpenWebUI down	PBS backups; migratable CTs
Randy	Backups + GPU NFS + Jellyfin	RAIDZ2 (2-disk/vdev); split HBAs
EX3400	Entire wired fabric	Dual PSU; UPS B
UPS A	Both R730s + Randy + DS4246	2200VA headroom; graceful shutdown
<code>pve1</code>	LAN DNS / ad-block	Secondary resolver <code>1.1.1.1</code>
<code>Corosync</code>	Loss of quorum	7 votes on stable VLAN-1 L2

15 Appendix — Live Port / Service Scan

Verified 2026-07-08 (`nmap -sT -sV, top-200`). `:9100`=node-exporter, `:3128`=Proxmox `pveproxy`. Proxmox `:8006`, PBS `:8007`, Jellyfin `:8096` are outside top-200 but known-live.

Host	Open ports	Identity
.1 OPNsense	53,80,443	Unbound DNS 1.23.1 + GUI
.2 UniFi USW	53,80,443,8080,8443	dnsmasq + UniFi
.31 Jarvis	22,111,3128, 8000 ,9100	SSH · Proxmox · llm_router · exporter
.50 EX3400	22,830	SSH + NETCONF
.100/.199 Ares	21,22,111,139,445,2049,9090	FTP · Samba · NFS · Cockpit
.112 Amazon Echo	1080,8888	Alexa; non-relaying SOCKS5 handshake (F-8)
.148 Homepage	22,8081	SSH · PeaNUT
.177 Pi-hole	22,53,80,443	dnsmasq pi-hole v2.92.2
.179 QuarkyLab	22,111,3128,9100	SSH · Proxmox · exporter
.180 UPS card	22,80,443	CyberPower
.181 NPM	22,80,81,443	OpenResty proxy
.182 Vaultwarden	22,80	Vaultwarden
.183 Observability	22,3000,8080,9100	Grafana · Scrutiny · exporter
.184 Wazuh	22,443	SIEM dashboard
.185 Open WebUI	22,8080	Chat UI
.186 Headscale	22,8080	Mesh VPN control
.187 Randy	22,111,2049,3128,9100	SSH · NFS · Proxmox · exporter (+8007/8096)
.201- .204 pve3/4/5/2	22,111,3128,9100	nodes (.204 also 443=step-ca)

Generated 2026-07-08 from live discovery reconciled against the NetFRAME Home-Lab vault.
Sanitized public edition. Sanitized edition suitable for the public repo generated alongside.